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```julia
# Block installing Julia packages needed to run this notebook.
begin
    using Plots
    using PlutoUI
    using PlutoTeachingTools
    using LaTeXStrings
    using Symbolics
    using NLsolve
    using HypertextLiteral: @htl, @htl_str
end
```

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Introduction

Opening remarks

Slides of the presentation: These summarise the following paragraphs and the content of the course in a short presentation. **TODO: Add slide link here**.

In modern practice **every scientist employs computers**. Some use them to symbolically derive equations for complex theories, others perform simulations, yet others analyse and visualise experimental results. In all these cases we encounter **mathematical problems**, which we need to **solve using numerical techniques**. The point of this lecture is to learn how to **formalise such numerical procedures** mathematically, **implement them** using the Julia programming language and **analyse using pen and paper** why some methods work better, some worse and some not at all.

You might think learning about this is a bit of an overkill and probably not very useful for your future studies. But let us look at a few examples, which give an overview what we will study in more depth throughout the course.

Interpolating data

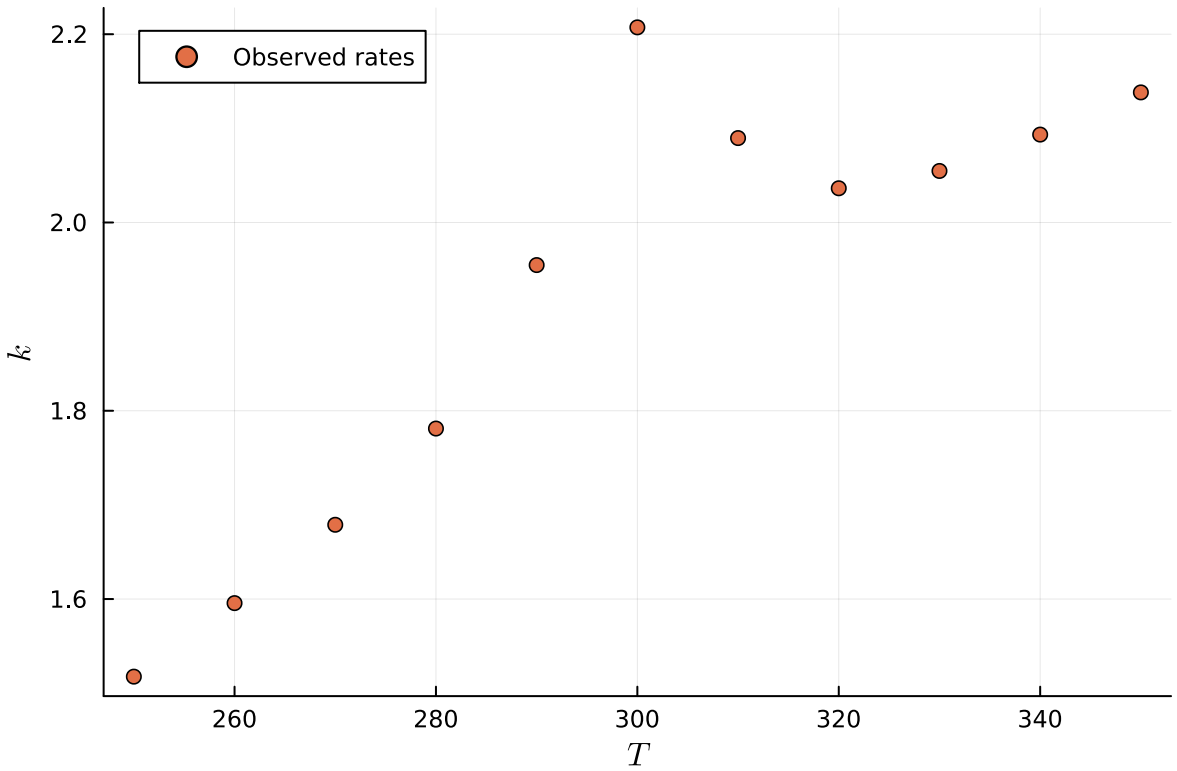
We start with the following problem: Imagine we did some measurement of the rate of a chemical reation at different temperatures, let's say

```julia
data = """
# Temperature(K)  Rate(1/s)
  250.0           1.51756
  260.0           1.59566
  270.0           1.67888
  280.0           1.78110
  290.0           1.95476
  300.0           2.20728
  310.0           2.08967
  320.0           2.03635
  330.0           2.05474
  340.0           2.09338
  350.0           2.13819
""";
```

Now we are curious about the rates at another temperature, say $255K$. The basic laws of thermodynamics and chemical kinetics tell us that the rate should be a *continuous function* of the temperature. For example Arrhenius' law

$$k(T) = A \exp\left(-\frac{E_A}{k_B T}\right) \tag{4}$$

establishes such a relationship, where A, E_A and k_B are some constants. Note, that in this case the behaviour is more complicated as is immediately obvious from a plot of the observed data:



Still, it seems reasonable that we should be able to **determine some continuous function f from the observed data**, which **establishes a good model** for the relationship $k = f(T)$. Albeit $255K$ has not been measured, we can thus evaluate $f(255K)$ and get an estimate for the rate at this temperature. Since the new temperature $255K$ islocated *within* the observed data range (i.e. here $250K$ to $350K$) we call this procedure **interpolation**.

In this lecture we will discuss some common interpolation techniques. In particular we will develop the **mathematical formalism** for these techniques and — most importantly — we will use this formalism to understand conditions when these methods work and **when these methods fail**.

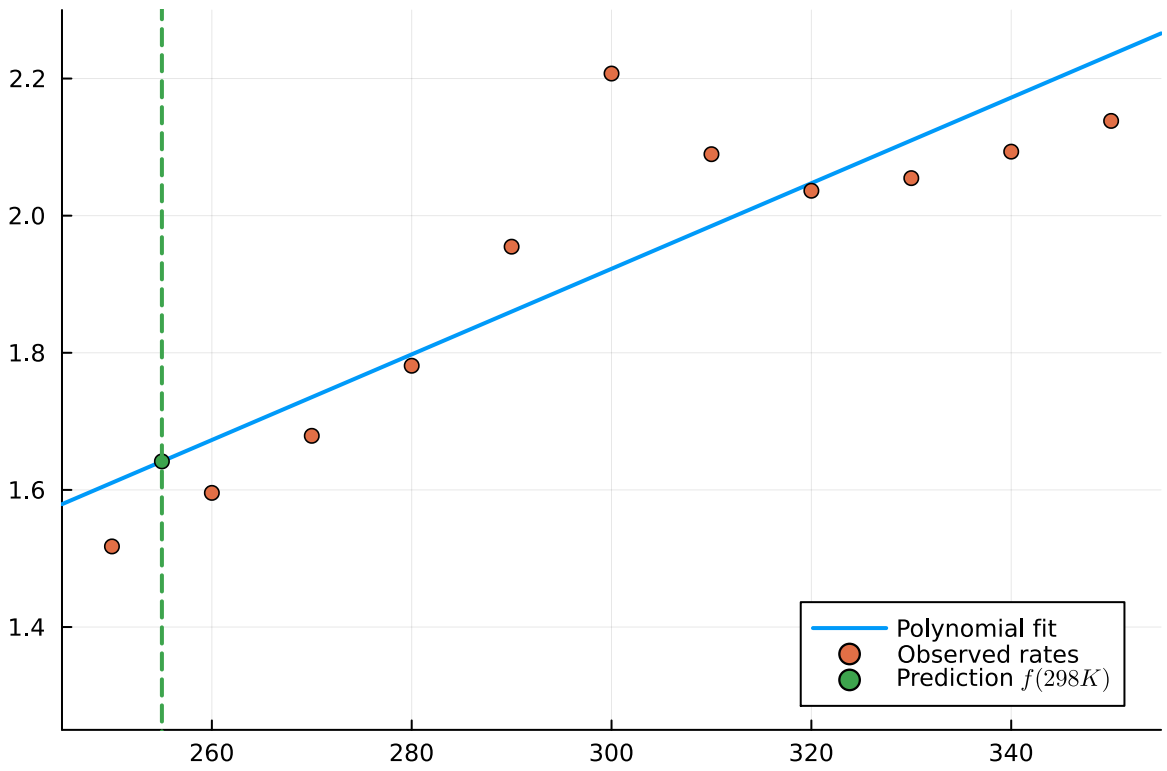
In fact our example here is already a little tricky. Let us illustrate this point for **polynomial interpolation**, one particularly frequent technique. Without going into details for now, one basic idea is to fit a polynomial of degree N

$$k(T) = \sum_{n=1}^{N} \alpha_n T^n$$

to the data. By some procedure discussed later we thus determine the unknown polynomial coefficients α_n, such that the polynomial best matches our observations and use that to determine $f(255K)$.

Let's see the result of such a fit. The slider let's you play with the polynomial order:

- N = 1



Clearly, the quality of such a fit depends strongly on the polynomial order. Perhaps suprising is, however, that **higher degree is not necessarily better**. We will discuss why this is.

Related questions we will discuss:

- How can I fit a polynomial to data points ?

- How accurate can I expect such a polynomial fit to be ?
- Can I choose an optimal polynomial order to obtain the most accurate answer ?
- If I have control over the strategy to acquire data (e.g. how to design my lab experiment), can I have an influence on the accuracy of such interpolations ?

Taking derivatives

Derivatices characterise the change of a quantity of interest. E.g. the acceleration as a derivative of the velocity indeed characterises how the velocity changes over time.

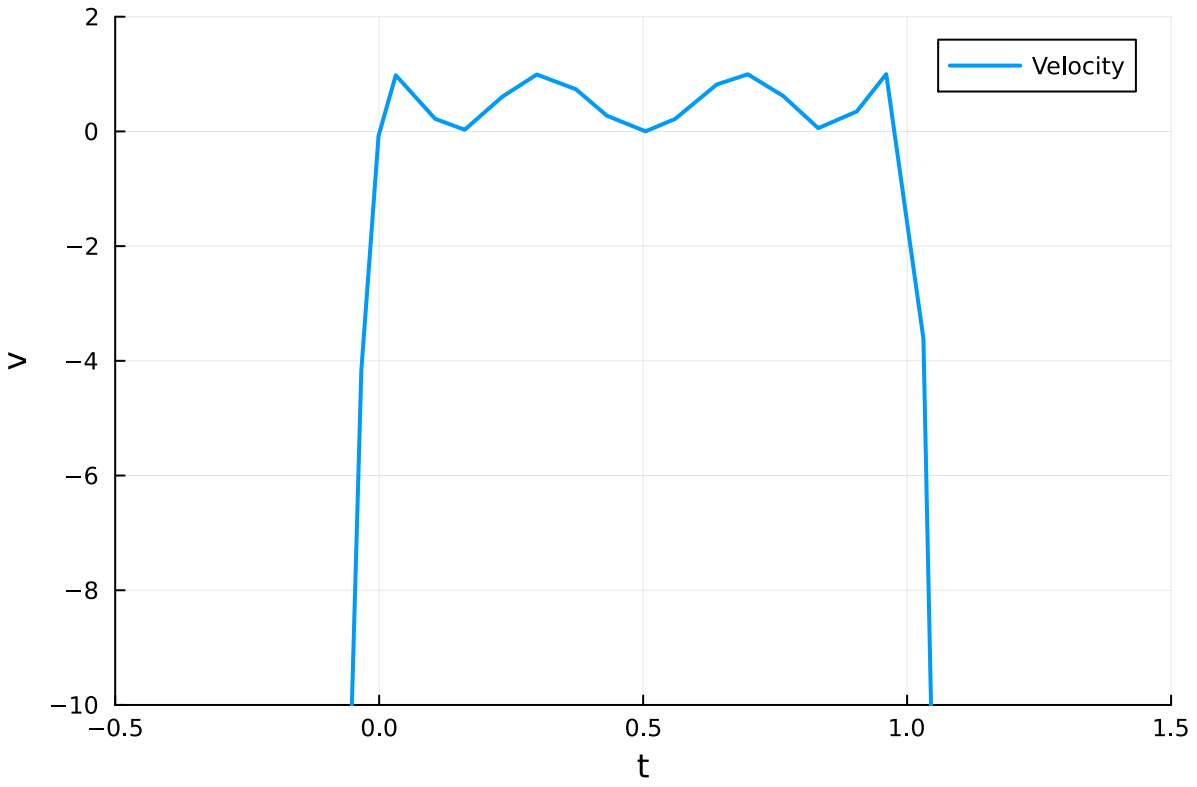
In a nutshel understanding changes of a materials or any other physical system as we interact with it, is the essential goal of pretty much every field of science and engineering.

As a result derivatives are everywhere. But computing derivatives by hand is not always easy. To see this consider the **innocent-looking velocity function**.

```
v (generic function with 1 method)
```

```julia
v(t) = 64t * (1 - t) * (1 - 2t)^2 * (1 - 8t + 8t^2)^2
```

We can plot it to get an idea:




```julia
let
    p = plot(v;
             ylims=(-10, 2),      # Limit on y axis
             xlims=(-0.5, 1.5),   # Limit on x axis
             linewidth=2,         # Width of the blue line
             label="Velocity",    # Label of the graph
             xlabel="t",
             ylabel="v")
end
```

Clearly between 0 and 1 the function changes quite rapidly. To investigate the **acceleration** there **we take the derivative**.

This can be done by hand, but instead we will employ a Julia package (namely `Symbolics`) to take the derivative for us. The result is:

$$64\left(1-8t+8t^2\right)^2(1-2t)^2\,(1-t)-64\left(1-8t+8t^2\right)^2(1-2t)^2t-256\left(1-8t+8t^2\right)^2t\,(1$$

```julia
let
    @variables t           # Define a variable
    dt = Differential(t)   # Differentiating wrt. t
    dv_dt = dt( v(t) )     # Compute dv / dt
    expand_derivatives(dv_dt)
end
```

Clearly far from a handy expression and not the kind of derivative one wants to compute by hand.

So let us **compute this derivative numerically** instead. An idea to do so goes back to the definition of the derivative $v'(t)$ as the limit $h \to 0$ of the slope of secants over an intervall $[t, t+h]$, i.e.

$$v'(t) = \lim_{h \to 0} \frac{v(t+h) - v(t)}{h}.$$

A natural idea is to not fully take the limit, i.e. to take a small $h > 0$ and approximate

$$v'(x) \approx \frac{v(t+h) - v(t)}{h}.$$

The expectation is that as we take smaller and **smaller values for h**, this **converges to the exact derivatives**.

So let's try this at the point `t0 = 0.2` for various values for `h`

```
h_values =
▸[1.0e-15, 3.16228e-15, 1.0e-14, 3.16228e-14, 1.0e-13, 3.16228e-13, 1.0e-12, 3.16228e-12,
```

```
h_values = 10 .^ (-15:0.5:1)
```

We compute the derivatives at each value for h ...

```
derivatives =
▸[8.99281, 9.0755, 9.05942, 9.06497, 9.06719, 9.06567, 9.06603, 9.06608, 9.06609, 9.06609,
```

```
derivatives = [ ( v(t0 + h) - v(t0) )/h for h in h_values ]
```

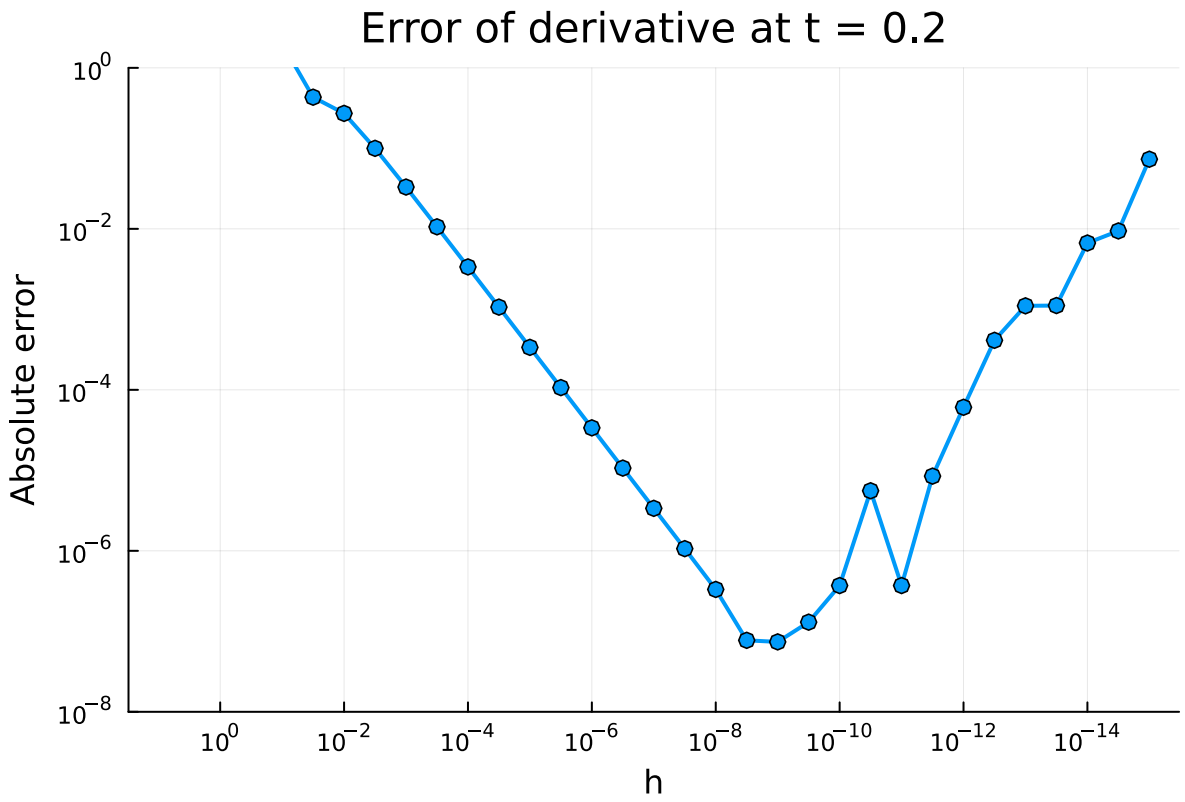
... and compute the error against a reference value:

```
▸[0.0732799, 0.00941812, 0.00666652, 0.00111438, 0.00110504, 0.000412217, 6.06921e-5, 8.47
```

```
begin
    using ForwardDiff # Package for computing derivatives of Julia functions
    reference = ForwardDiff.derivative(v, t0)

    errors = [abs(d - reference) for d in derivatives]
end
```

Finally we plot the errors on a log-log plot:




```
let
    plot(h_values, errors;
        xaxis=:log, yaxis=:log,
        xflip=true,  # Flip order of x axis
        ylims=(1e-8, 1),
        mark=:o,     # Mark all points by circle
        xlabel="h",
        ylabel="Absolute error",
        linewidth=2,
        label="",
        xticks=10.0 .^ (-16:2:0),
        title="Error of derivative at t = $t0",
    )
end
```

We observe that while indeed initially the error decreases as h decreases at some point it starts to increase again with **results worse and worse**.

With this slider you can check this is indeed the case for pretty much all values of t where we take the derivative numerically:

t0 = 0.2

Related questions we will discuss:

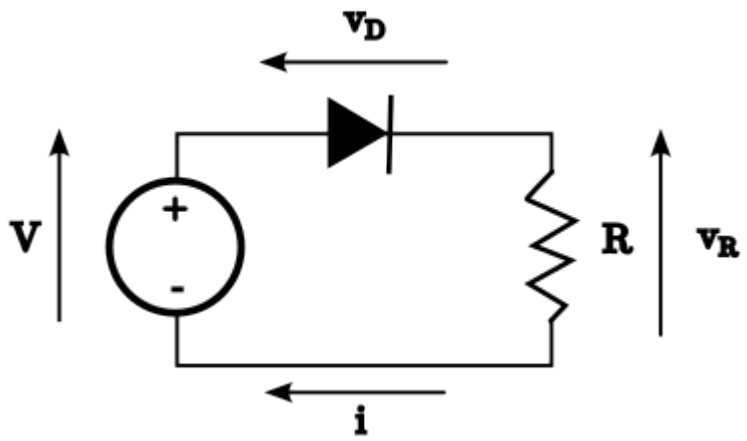
- There seems to be some optimal value for h. Is it independent of the function to be differentiated ?
- Here seems to be some minimal error we can achieve. Are there more accurate derivative formulas ?
- The convergence plot as h decreases seems to have the same slope (convergence rate) for all points t. Does this slope depend on f ? How can we reach faster convergence ?

Solving differential equations

TODO Some motivating ODE example

Modelling electric circuits

Let us consider the simple Diode model circuit



which consists of

- Some voltage source, generating a voltage V
- A resistor R with the linear relationship

$$v_R = Ri \tag{1}$$

  between the current i and its voltage v_R.
- A diode for which we will take the standard Shockley diode model, i.e. the equation

$$i = i_0(e^{v_D/v_0} - 1) \tag{2}$$

between the diode's current i and voltage v_D. v_0 and i_0 are parameters, which characterise the diode.

We now want to solve this problem numerically. This notebook already defines variables for the parameters R, V, i_0 and v_0. This is done using sliders, which you find further down this notebook. Currently these variables have the values

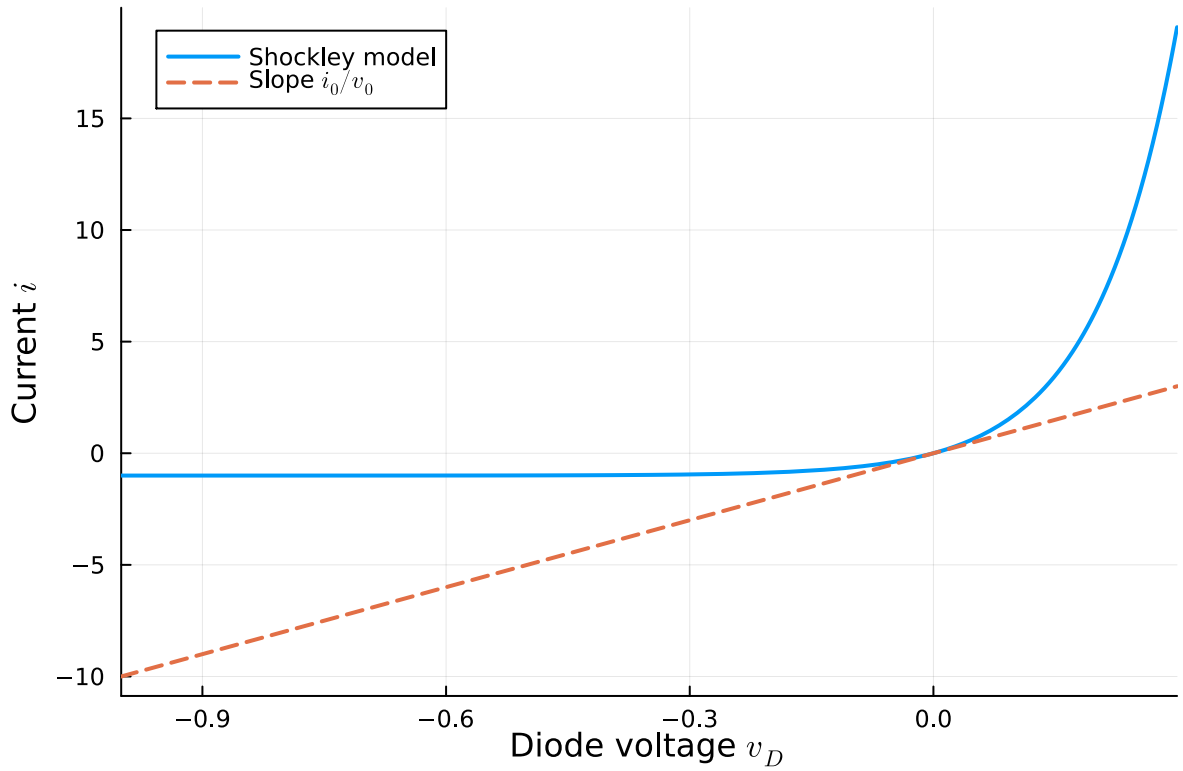
- `R = 1.0`
- `V = 1.0`
- `i0 = 1.0`
- `v0 = 0.1`

In Julia code we can thus define the Shockley diode relationship, mapping v_D to i as follows

```
shockley_diode (generic function with 1 method)
```

```julia
function shockley_diode(vD)
    i0 * (exp(vD/v0) - 1)
end
```

Graphically for above parameters this looks like




```
let
    p = plot(shockley_diode, xlim=(-1, 0.3);
             label="Shockley model",
             xlabel=L"Diode voltage $v_D$",
             ylabel=L"Current $i$",
             linewidth=2)
    plot!(p, v -> (i0/v0) * v;
          label=L"Slope $i_0 / v_0$",
          linestyle=:dash,
          linewidth=2)
end
```

Our goal is to model this circuit, i.e. understand its current i the voltages across the circuit elements.

First we balance the supplied voltage V across the circuit. Using equations (1) and (2) this leads to the following:

$$V = v_R + v_D = Ri + v_D = Ri_0\left(e^{v_D/v_0} - 1\right) + v_D \tag{3}$$

We introduce the function

$$f(v_D) = Ri_0\left(e^{v_D/v_0} - 1\right) + v_D - V,$$

which makes (3) equivalent to seeking a diode voltage v_D such that $f(v_D) = 0$ — a **root finding problem**.

However, since f is not just a simple polynomial, but a **non-linear function** there is no explicit analytic formula for finding its solution. We are thus forced to employ numerical methods.

As we will see in the next lectures there are a number of numerical techniques to solve such nonlinear problems $f(x) = 0$. However, similar to polynomial fitting, none of them always just work. In this case we are lucky and can actually directly defer to a standard package of the Julia programming language, called `NLsolve` to solve our problem:

```
▸ [0.0659611]
```

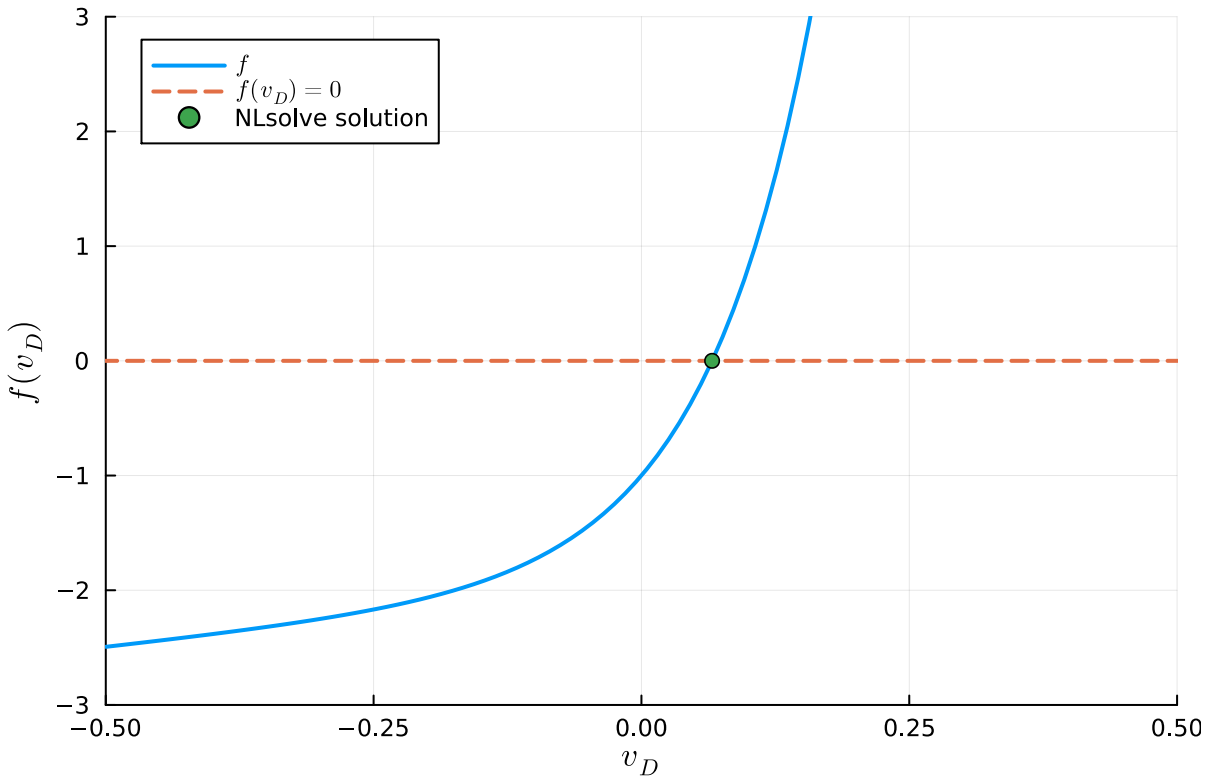
```julia
begin
    # Define our function
    f(vD) = R*i0*(exp(vD/v0) - 1) + vD - V

    # Solve it with nlsolve
    z = nlsolve(v -> f(v[1]), [0.0]).zero
end
```

which for the parameters

- i0 = 1.0
- v0 = 0.1
- R = 1.0
- V = 1.0

tells us the root is located at $v_D \simeq 0.066$. We can also check this graphically:




```
let
    vs = range(-0.5, 0.5; length=100)
    plot(vs, f.(vs); xlims=(-0.5, 0.5), ylims=(-3, 3), legend=:topleft,
            label=L"f", linewidth=2, xlabel=L"v_D", ylabel=L"f(v_D)")
    hline!([0], linestyle=:dash, linewidth=2, label=L"f(v_D) = 0")
    scatter!(z, [0], label="NLsolve solution")
end
```

Given this nice `NLsolve` package, which seems to do the trick automatically, you might wonder **why we even bother** discussing, analysing and implementing such techniques at all.

Well, unfortunately **for many realistic numerical settings** (e.g. the quantum-chemical simulation of materials) even **standard packages** such as NLsolve are **no longer able to automatically figure out the best solution strategy**. As a result developing some intuition for the required numerical procedure.

Even if you now might think, that your heart beats for **experimental research** and you will never need numerics, keep in mind that **all data analysis** uses procedures **based on such techniques**. **Even commercial packages** for experimental post-processing rely heavily on the procedures we will cover and **sometimes get it wrong}}. See for example this report on Microsoft Excel). This **can and has compromised scientific fundings in the past**. Therefore it's **best to be prepared**, so you can judge what is wrong: The experiment or the numerics.

Related questions we will discuss:

- How can I understand whether an obtained numerical answer is credible ?
- What techniques based on visualisation or plotting help me to understand the accuracy of a numerical algorithm ?
- How can I overcome numerical issues ?
- What are numerically stable techniques for interpolation, basic data analysis or solving standard scientific problems ?
- How can I understand and evaluate the speed of convergence of an algorithm and improve it even further ?

Numerical analysis

1. Introduction
2. The Julia programming language
3. Revision and preliminaries
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5. Interpolation
6. Direct methods for linear systems
7. Iterative methods for linear systems
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9. Numerical integration
10. Numerical differentiation
11. Initial value problems
12. Boundary value problems